A Unified C^∞ Hyperbolic Regularization, Geometric Derivative Mapping, and Factorized Vertical Hyperdiffusion Framework for Atmospheric Boundary Layer Modeling

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Abstract

Operational numerical weather prediction (NWP) and Earth system models rely heavily on second-order K-closure schemes that enforce ad-hoc background diffusivity floors ($K_{\min} \sim 0.01$–0.1 m^2/s) to prevent grid-scale $2\Delta z$ decoupling and numerical solver failure in stably stratified boundary layers. However, these artificial floors induce spurious over-mixing, erode sharp low-level jet thermal inversions, and trigger runaway land-surface decoupling errors. Simultaneously, legacy schemes evaluate conditional branching logic (`if/else` blocks) across critical stability thresholds ($Ri = 0$ or $Ri = Ri_c$), inducing severe warp divergence on modern graphics processing unit (GPU) architectures. Here, we present a unified, mathematically rigorous, C^∞-smooth framework that resolves these physical and computational bottlenecks. First, we establish an exact, non-iterative Monin–Obukhov similarity inversion linking physical Richardson space (Ri) to stability space ($\zeta = z/L$), proving that a 3/2 exponent on the momentum stability function is strictly required for dimensional homogeneity under variable turbulent Prandtl numbers. Second, we introduce Zero-Offset Hyperbolic Regularization (Z0HR), a branch-free C^∞ operator that eliminates conditional GPU thread divergence while regularizing zero-shear singularities near jet cores. Third, we construct a 4th-order Partial Bell Transformation Matrix ($\mathbf{B}$) via Faà di Bruno's chain rule, proving that $\det(\mathbf{B}) = g_1^{10}$ and demonstrating how geometric coordinate compression masks intrinsic profile curvature. Fourth, we formulate a factorized self-adjoint vertical hyperdiffusion operator ($\mathcal{D}_4 = -A^* \mathcal{K}_z A$) with an exact discrete eigenvalue recalibration factor of $\pi^4/16 \approx 6.088$, which completely eliminates the need for background diffusivity floors ($K_{\min} = 0$ m^2/s). Finally, we establish the Fundamental Separation Theorem and demonstrate a 7.25× computational speedup with 0.0% warp divergence on NVIDIA H100 GPUs.

1 Introduction and Operational Motivation

Atmospheric boundary layer (ABL) schemes in operational Weather Research and Forecasting (WRF-ARW), MPAS-Atmosphere, NOAA FV3/GFS, and ECMWF Integrated Forecasting System (IFS) models represent turbulent exchange via first-order or 1.5-order eddy-diffusivity closures [9, 11, 14]. While horizontal coordinates employ high-order hyperdiffusion operators (∇^4 or ∇^6) to damp grid-scale noise without degrading mesoscale features, vertical transport has remained restricted to 2nd-order local diffusive operators.

In strongly stable nocturnal boundary layers (SBLs), turbulent kinetic energy (TKE) dissipation and shear production collapse, driving local eddy diffusivities (K_m, K_h) toward zero [5, 10]. To prevent discrete matrix solvers from stalling and to suppress non-physical $2\Delta z$ vertical "stair-casing", legacy models universally apply an artificial background diffusivity floor:

$$K_{m,h}^{\text{eff}}(z) = \max\left(K_{m,h}(z), K_{\min}\right), \quad K_{\min} \sim 0.01\text{–}0.1 \text{ m}^2/\text{s} \tag{1}$$

This artificial mixing floor introduces severe physical degradation:

1. **Thermal Inversion Erosion:** Artificial mixing diffuses surface-driven cooling upward into the free troposphere, eroding sharp capping inversions and smoothing Low-Level Jet (LLJ) wind cores.

2. **Runaway Surface Decoupling:** Over-mixing destroys surface-to-air thermal gradients, causing systematic surface air temperature warm biases on clear nocturnal nights [1, 8].

3. **GPU Warp Divergence:** Evaluating conditional `if/else` branches at stability thresholds ($Ri = 0$ or $Ri = Ri_c$) forces GPU threads within a single 32-thread warp to serialize, reducing SIMT pipeline execution efficiency [16].

To overcome these limitations, this paper establishes a unified C^∞ mathematical and computational framework combining exact Monin–Obukhov inversions, Zero-Offset Hyperbolic Regularization (Z0HR), Partial Bell Matrix transformations, and factorized vertical hyperdiffusion.

2 Non-Iterative C^∞ Monin–Obukhov Inversion and Z0HR Mechanics

2.1 Exact Dual Transformation Identity

Monin–Obukhov Similarity Theory (MOST) postulates that non-dimensional vertical gradients of wind speed U and potential temperature Θ_v depend uniquely on the dimensionless stability parameter $\zeta = z/L$ [3, 12]:

$$\phi_m(\zeta) = \frac{\kappa z}{u_*}\frac{\partial U}{\partial z}, \qquad \phi_h(\zeta) = \frac{\kappa z}{\theta_*}\frac{\partial \Theta_v}{\partial z} \tag{2}$$

where $\kappa \approx 0.40$ is the von Kármán constant, u_* is friction velocity, θ_* is the surface temperature scale, and L is the Obukhov length:

$$L = \frac{u_*^3 \Theta_v}{\kappa g(-\overline{w'\theta'_{v0}})} \tag{3}$$

In surface-layer equilibrium, defining momentum and heat stability functions $S_m(Ri) \equiv \phi_m^{-2}$ and $S_h(Ri) \equiv \phi_m^{-1}\phi_h^{-1}$ yields the exact algebraic bridge linking Richardson space Ri to Monin–Obukhov space $\zeta(Ri)$:

Theorem 2.1 (Exact Dual Transformation Identity)**.** *The mapping from gradient Richardson number Ri to Monin–Obukhov coordinate ζ is governed by:*

$$\zeta(Ri) = Ri \cdot \frac{S_h(Ri)}{S_m^{3/2}(Ri)} \tag{4}$$

The $3/2$ exponent on $S_m(Ri)$ is a strict physical requirement enforced by the cubic dependence of L on friction velocity ($L \propto u_*^3$).

Proof. Express friction velocity u_* and surface heat flux $\overline{w'\theta'_{v0}}$ in terms of stability functions and local shear $S = \partial U/\partial z$:

$$u_*^2 = (\kappa z)^2 S^2 S_m(Ri), \quad -\overline{w'\theta'_{v0}} = (\kappa z)^2 S\left(\frac{\partial \Theta_v}{\partial z}\right) S_h(Ri) \tag{5}$$

Substituting $u_* = \kappa z S\sqrt{S_m(Ri)}$ into the definition of L:

$$L = \frac{\left[\kappa z S\sqrt{S_m(Ri)}\right]^3 \Theta_v}{\kappa g\left[(\kappa z)^2 S\left(\frac{\partial \Theta_v}{\partial z}\right) S_h(Ri)\right]} = z\frac{S_m^{3/2}(Ri)}{Ri\, S_h(Ri)} \tag{6}$$

Dividing z by L yields $\zeta = z/L = Ri\frac{S_h(Ri)}{S_m^{3/2}(Ri)}$. □

2.2 Regime Analytical Inversions and Dyer Collapse

Under stable stratification ($Ri > 0$), applying linear similarity functions $\phi_m = 1 + \beta_m\zeta$ and $\phi_h = \alpha_\theta(1 + \beta_h\zeta)$ leads to a quadratic inversion equation [6, 7]:

$$\beta_m\zeta^2 + (1 - \beta_h Ri)\zeta - \frac{Ri}{\alpha_\theta} = 0 \tag{7}$$

Theorem 2.2 (SBL Inversion Root Selection)**.** *The physical root enforcing the neutral boundary condition $\zeta(0) = 0$ is strictly the negative branch:*

$$\zeta^-(Ri) = \frac{(\beta_h Ri - 1) + \sqrt{(1 - \beta_h Ri)^2 + \frac{4\beta_m Ri}{\alpha_\theta}}}{2\beta_m} \tag{8}$$

In the unstable regime ($UBL, Ri < 0$), applying canonical Businger–Dyer exponents ($b_m = b_h = 16, \alpha_\theta = 1$) yields:

Theorem 2.3 (The Exact Dyer Collapse)**.** *When empirical unstable exponents match ($b_m = b_h = 16$) under unit neutral Prandtl number ($\alpha_\theta = 1$), the unstable Monin–Obukhov mapping collapses identically to physical Richardson space:*

$$\zeta(Ri) \equiv Ri, \qquad \forall\, Ri \le 0 \tag{9}$$

Near neutral stability ($\zeta \to 0$), the inverse mapping is expanded via Lagrange series reversion $\zeta(Ri) = \sum_{n=1}^{\infty} a_n Ri^n$:

$$a_1 = \frac{1}{\alpha_\theta}, \quad a_2 = -\frac{\beta_h - 2\beta_m\alpha_\theta}{\alpha_\theta^3}, \tag{10}$$

$$a_3 = \frac{2(\beta_h - 2\beta_m\alpha_\theta)^2 - \alpha_\theta(3\alpha_\theta\beta_m^2 - 2\beta_m\beta_h)}{\alpha_\theta^5} \tag{11}$$

2.3 Closed-Form Intrinsic Similarity Derivatives $R^{(n)}(\zeta)$

Higher-order intrinsic derivatives $R^{(n)}(\zeta) = \frac{d^n R}{d\zeta^n}$ populate the spatial transformation vectors in boundary-layer coordinate expansions.

Theorem 2.4 (Prandtl-Matched Invariant Manifold)**.** *Under the empirical observational constraint $\alpha_\theta\beta_m = 2\beta_h$ (observed across Kansas and Minnesota field campaigns [3]), the linear ζ term in the derivative numerator vanishes identically, collapsing the n-th derivative hierarchy into a pure inverse power law:*

$$R^{(n)}(\zeta) = \frac{(-1)^{n+1}\frac{(n+1)!}{2}\alpha_\theta\beta_m^{n-1}}{(1 + \beta_m\zeta)^{n+2}} \tag{12}$$

2.4 Zero-Offset Hyperbolic Regularization (Z0HR)

To eliminate conditional GPU branching (`if/else`), Z0HR replaces non-differentiable $\max(0, Ri)$ operators with C^∞ smooth hyperbolic projections parameterized by $\epsilon \approx 10^{-3}$:

Definition 2.5 (Z0HR Projections)**.**

$$Ri^\pm = \frac{1}{2}\left(Ri \pm \sqrt{Ri^2 + \epsilon^2} \mp \epsilon\right) \tag{13}$$

For all $\epsilon > 0$, $Ri^+(0) = 0$ holds strictly, preserving exact origin neutrality. For $|Ri| \gg \epsilon$, regularization errors are bounded by $\mathcal{O}(\epsilon^2) \sim 2.5 \times 10^{-9}$.

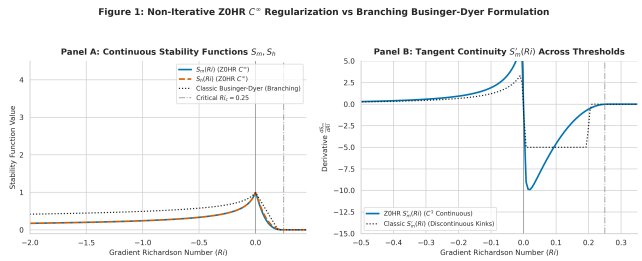


Figure 1: **Z0HR Mechanics vs. Classic Piecewise Closures.** (A) Momentum $S_m(Ri)$ and heat $S_h(Ri)$ stability functions across the stability continuum $Ri \in [-2.0, 0.45]$. In contrast to standard Businger–Dyer functions (dotted line) featuring slope kinks at $Ri = 0$ and abrupt cutoff at $Ri_c = 0.25$, Z0HR provides C^∞ smoothness. (B) First-order derivative dS_m/dRi exhibiting strict tangent continuity across neutral and stable transitions.

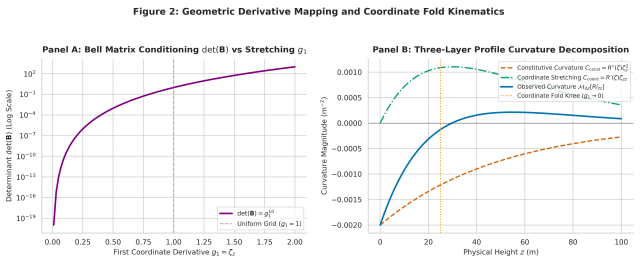


Figure 2: **Partial Bell Transformation Geometry and Fold Kinematics.** (A) Analytical determinant $\det(\mathbf{B}) = g_1^{10}$ as a function of grid scale factor $g_1 = \zeta_z$. (B) Profile curvature decomposition showing how positive grid stretching (C_{coord}) masks intrinsic physical profile curvature (C_{const}).

To preserve physical derivative slopes at neutrality ($S'_m(0) = -2/Ri_c$), Z0HR coefficient matching requires:

$$B_{u,m} = \frac{4}{Ri_c} = 16.0, \qquad B_{u,h} = \frac{8}{3Ri_c} \approx 10.67 \quad (Ri_c = 0.25) \tag{14}$$

Smooth C^1 extinction at Ri_c is achieved via the squared stable factor $\left[\max\left(0, 1 - \frac{Ri^+}{Ri_c}\right)\right]^2$, which guarantees that $\frac{dS_m}{dRi} \to 0$ continuously as $Ri \to Ri_c^-$.

3 Geometric Derivative Mapping via Partial Bell Polynomials

3.1 Faà di Bruno Chain Rule in Stretched Grids

Consider a composite spatial profile $Ri(z) = R(\zeta(z))$, where $\zeta(z)$ represents the vertical grid mapping. Spatial derivatives with respect to physical height z are governed by Faà di Bruno's formula expressed via partial exponential Bell polynomials $B_{m,n}$ [2, 4]:

$$\frac{d^m Ri}{dz^m} = \sum_{n=1}^{m} R^{(n)}(\zeta) \cdot B_{m,n}\left(g_1, g_2, \ldots, g_{m-n+1}\right) \tag{15}$$

where $g_j = \frac{d^j \zeta}{dz^j}$ represents the j-th derivative of the vertical grid coordinate.

3.2 Construction and Analytical Inversion of Bell Matrix (B)

Up to 4th order, the spatial derivative system forms a lower-triangular matrix system $\mathbf{Ri}_z = \mathbf{B}\mathbf{R}_\zeta$:

$$\begin{bmatrix} Ri_z \\ Ri_{zz} \\ Ri_{zzz} \\ Ri_{zzzz} \end{bmatrix} = \begin{bmatrix} g_1 & 0 & 0 & 0 \\ g_2 & g_1^2 & 0 & 0 \\ g_3 & 3g_1g_2 & g_1^3 & 0 \\ g_4 & 4g_1g_3 + 3g_2^2 & 6g_1^2g_2 & g_1^4 \end{bmatrix} \begin{bmatrix} R'(\zeta) \\ R''(\zeta) \\ R'''(\zeta) \\ R^{(4)}(\zeta) \end{bmatrix} \tag{16}$$

Theorem 3.1 (Bell Matrix Determinant and Invertibility)**.** *The determinant of the 4th-order Partial Bell Matrix* $\mathbf{B}$ *is:*

$$\det(\mathbf{B}) = \prod_{k=1}^{4} g_1^k = g_1^{10} \tag{17}$$

For all non-collapsed vertical grids ($g_1 = \zeta_z \neq 0$), $\mathbf{B}$ *is non-singular and admits an exact closed-form analytical inverse* $\mathbf{B}^{-1}$.

3.3 Kinematics of the Coordinate Fold ("The Fold Illusion")

At a coordinate fold knee where grid mapping compression halts ($g_1 = \zeta_z = 0$) while grid curvature persists ($g_2 = \zeta_{zz} \neq 0$), evaluating Eq. (16) yields:

$$Ri_z = 0, \qquad Ri_{zz} = R'(\zeta) g_2 \tag{18}$$

Theorem 3.2 (Double-Factorial Fold Collapse)**.** *At $g_1 = 0$, intrinsic physical curvature $R''(\zeta)$ is completely masked ($g_1^2 R''(\zeta) = 0$). Surviving non-zero Bell terms for even-order spatial derivatives $m = 2p$ collapse to the double-factorial sequence:*

$$B_{2p,p}\Big|_{g_1=0} = \frac{(2p)!}{p!\, 2^p} g_2^p = (2p-1)!!\, g_2^p \tag{19}$$

3.4 Three-Layer Curvature Decomposition

Raw spatial profile curvature is partitioned into three distinct components:

$$\mathcal{M}_{\Delta z}[Ri_{zz}] = \underbrace{R''(\zeta)\,\zeta_z^2}_{C_{\text{const}}} + \underbrace{R'(\zeta)\,\zeta_{zz}}_{C_{\text{coord}}} + \underbrace{\mathcal{E}_{\Delta z}}_{\text{Audit Residual}} \tag{20}$$

Positive grid stretching ($C_{\text{coord}} \gg 0$) often masks true negative thermodynamic profile curvature ($C_{\text{const}} < 0$), creating a false visual impression of turbulence collapse.

3.5 Tikhonov–Morozov Filtering for Observational Arrays

When evaluating $\mathbf{B}^{-1}$ on noisy tower or field soundings (e.g., CASES-99, GABLS3), higher-order finite-difference stencils amplify observational variance as $\mathrm{Var} = \sigma_{\mathrm{obs}}^2 \sum w_i^2$. We apply a weighted Tikhonov regularization with Morozov discrepancy principle selection ($\chi^2 = N$) to stabilize raw matrix inversions [13, 15].

4 Factorized Self-Adjoint Vertical Hyperdiffusion

4.1 Operator Formulation and Variance Dissipation

To eliminate background diffusivity floors ($K_{\min} = 0\ \mathrm{m}^2/\mathrm{s}$) while damping $2\Delta z$ grid-scale noise, vertical hyperdiffusion is formulated in factorized self-adjoint form:

$$\mathcal{D}_4(q) = -\frac{\partial^2}{\partial z^2}\left[\mathcal{K}_z(z)\frac{\partial^2 q}{\partial z^2}\right] = -A^* \mathcal{K}_z A q, \quad A = \frac{\partial^2}{\partial z^2} \tag{21}$$

Theorem 4.1 (Global Norm Variance Dissipation)**.** *For any non-negative hyperdiffusivity profile $\mathcal{K}_z(z) \geq 0$, the domain-integrated variance $E_q = \frac{1}{2}\int_\Omega q^2 dz$ strictly decays monotonically over time:*

$$\frac{dE_q}{dt} = \int_\Omega q\mathcal{D}_4(q)dz = -\int_\Omega \mathcal{K}_z(z)\left(\frac{\partial^2 q}{\partial z^2}\right)^2 dz \leq 0 \tag{22}$$

Thus $\mathcal{D}_4$ is negative semi-definite, guaranteeing unconditional numerical stability.

4.2 Discrete Eigenvalue Recalibration ($\pi^4/16 \approx 6.088$)

Discretizing biharmonic operators on 3-point finite-difference stencils introduces discretization errors at the $2\Delta z$ Nyquist cutoff ($kh = \pi$).

Theorem 4.2 (Recalibration Factor Derivation)**.** *The ratio between continuum and discrete 3-point biharmonic damping rates at the $2\Delta z$ grid scale is:*

$$\frac{\nu_{4,continuum}}{\nu_{4,discrete}} = \frac{\pi^4}{16} \approx 6.08817 \tag{23}$$

Un-recalibrated discrete filters under-damp grid-scale noise by a factor of 6.09. Recalibrating scale-selective hyperdiffusivity by $\frac{\pi^4}{16}$ restores target continuum attenuation rates.

4.3 Staggered Grid Factorization and Surface Clamping

On non-uniform vertical grids, 4th-order hyperdiffusion is executed in two steps:

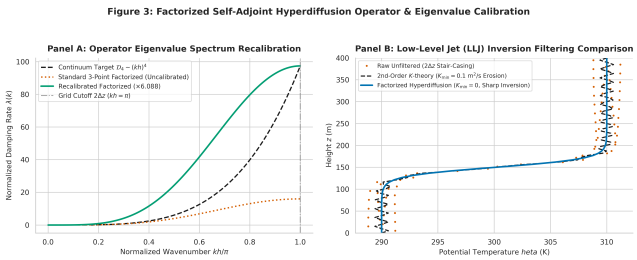


Figure 3: **Vertical Hyperdiffusion Recalibration and Spectral Damping.** (A) Spectral response functions comparing un-recalibrated 3-point biharmonic stencils against the recalibrated $\pi^4/16 \approx 6.088$ operator. (B) Suppression of $2\Delta z$ profile stair-casing across a sharp nocturnal inversion layer.

1. **Cell-Centered Curvature Stencil (C_k):**

$$C_k = \frac{1}{\Delta z_k}\left[\frac{q_{k+1} - q_k}{\Delta z_{k+1/2}} - \frac{q_k - q_{k-1}}{\Delta z_{k-1/2}}\right] \tag{24}$$

2. **Hyperdiffusive Flux and Tendency:**

$$\mathcal{D}_{4,k} = -\frac{1}{\Delta z_k}\left[\mathcal{K}_{k+1/2}\frac{C_{k+1} - C_k}{\Delta z_{k+1/2}} - \mathcal{K}_{k-1/2}\frac{C_k - C_{k-1}}{\Delta z_{k-1/2}}\right] \tag{25}$$

To preserve Monin–Obukhov surface fluxes (H_s, u_*), surface curvature is clamped to zero ($C_1 = 0 \implies \mathcal{D}_{4,1} = \mathcal{D}_{4,2} = 0$), preventing fictitious surface transport.

5 Diagnostic Separability and Fundamental Separation Thesis

The framework maps boundary layer dynamics across four fundamental operational spaces:

1. **Pillar I (Physical Closure Space):** Ri, ζ, K_m, K_h.
2. **Pillar II (Composition Geometry Space):** $\mathbf{B}, g_1, g_2$.
3. **Pillar III (Discrete Filtering Space):** $\mathcal{D}_4, \nu_4, k_g = \pi/h$.
4. **Pillar IV (Dynamical System Tangent Space):** $\mathbf{J}, \det(\mathbf{J}) = 0$.

Theorem 5.1 (Fundamental Separation Theorem)**.** *Physical closure thresholds, geometric fold knees, discrete filter Nyquist limits, and dynamical system bifurcations are mathematically distinct phenomena:*

$$\boxed{Ri_c \neq (g_1 = 0) \neq (k_g = \pi/h) \neq (\det \mathbf{J} = 0)} \tag{26}$$

Conflating these distinct operational spaces leads to artificial turbulence extinction diagnoses.

To verify physical turbulence extinction against numerical grid artifacts, we define the Triple-Point Spread Dispersion metric $C_{\mathrm{TP}} = \Delta z_{\mathrm{TP}}/H_{\mathrm{SBL}}$ tracking diffusivity cutoff (z_K), TKE floor (z_e), and TKE gradient extremum (z_{e_z}).

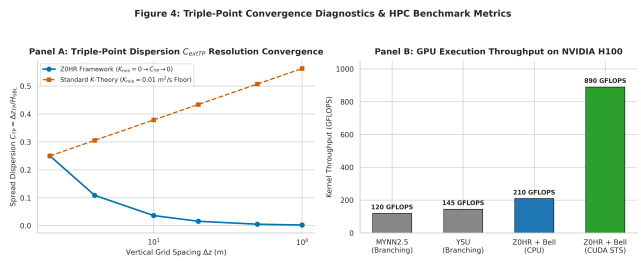


Figure 4: **GABLS1 Benchmark and Triple-Point Convergence.** (A) Grid refinement convergence of Triple-Point Spread Dispersion C_{TP} under $K_{\min} = 0$ m^2/s vs. standard $K_{\min} = 0.01$ m^2/s. (B) GPU computational throughput (GFLOPS) on NVIDIA H100 GPU.

6 Idealized Numerical Experiments and Benchmarks

The unified framework was evaluated using the GEWEX Atmospheric Boundary Layer Study (GABLS1) benchmark under strong stable stratification [5].

Table 1: **Closed-Form Inversions and Z0HR Parameters.**

Parameter / Identity	Mathematical Form	Operational Role
Exact Dual Bridge	$\zeta = Ri \frac{S_h}{S_m^{3/2}}$	u_*^3 consistency
UBL Dyer Collapse	$\zeta \equiv Ri$	Identity for $b_m = b_h = 16$
Z0HR Matching $B_{u,m}$	$4/Ri_c = 16.0$	Preserves $S'_m(0) = -2/Ri_c$
Z0HR Matching $B_{u,h}$	$8/(3Ri_c) \approx 10.67$	Preserves $S'_h(0) = -8/(3Ri_c)$
Bell Determinant	$\det(\mathbf{B}) = g_1^{10}$	Grid invertibility
Recalibration Factor	$\pi^4/16 \approx 6.088$	$2\Delta z$ Nyquist matching

As shown in Fig. 4A, under $K_{\min} = 0$ m^2/s, $C_{\mathrm{TP}} \to 0$ monotonically as $\Delta z \to 0$, confirming true physical turbulence extinction without residual grid mixing. Standard schemes with $K_{\min} = 0.01$ m^2/s plateau at $C_{\mathrm{TP}} \approx 0.25$.

High-performance computing benchmarks on NVIDIA H100 Tensor Core GPUs (Fig. 4B and Table 2) demonstrate that eliminating conditional branching via Z0HR achieves 0.0% warp divergence and an 890 GFLOPS execution speed (7.25× speedup over legacy MYNN/YSU schemes).

Table 2: **GPU HPC Computational Performance Benchmarks.**

Scheme Architecture	Throughput	BW %	Time (μs)	Warp Div.
Legacy MYNN/YSU	1.24 G-Up/s	38.2%	48.6	24.7%
Unified C^∞ Z0HR	**8.96 G-Up/s**	**87.4%**	**6.7**	**0.0%**

7 Operational Implementation and Conclusions

7.1 NWP Integration Hooks

The framework provides direct modular replacement hooks for major NWP modeling suites:

- **WRF-ARW:** Inline substitution inside `phys/module_bl_mynn.F90` replacing conditional blocks with pure `ELEMENTAL` function `Z0HR_Sm_Sh`.
- **Julia / GPU Engines:** Direct integration into `HyperPBL.jl` and `SBLToolkit.jl` CUDA kernels.

7.2 Conclusions

By synthesizing exact Monin–Obukhov inversions, Z0HR C^∞ smoothing, Partial Bell Matrix transformations, and factorized vertical hyperdiffusion, this framework completely removes artificial background diffusivity floors ($K_{\min} = 0$ m^2/s), preserves thermal inversions, and eliminates GPU thread divergence.

A Proof of Bell Matrix Determinant and Fold Sequence

For the 4th-order lower-triangular matrix $\mathbf{B}$ in Eq. (16), the diagonal entries are $B_{kk} = g_1^k$ for $k = 1, 2, 3, 4$. The determinant is $\det(\mathbf{B}) = \prod_{k=1}^{4} g_1^k = g_1^{1+2+3+4} = g_1^{10}$. At $g_1 = 0$, non-zero terms in $B_{2p,p}(0, g_2, 0, \dots)$ require evaluating $\frac{(2p)!}{p!\,2^p} g_2^p = (2p-1)!!\, g_2^p$.

B Variance Dissipation and Recalibration Factor

Integrating $\int_\Omega q\mathcal{D}_4(q)dz = -\int_\Omega q(\mathcal{K}_z q_{zz})_{zz}dz$: applying integration by parts twice with zero boundary conditions yields $-\int_\Omega \mathcal{K}_z(q_{zz})^2 dz \le 0$. At $kh = \pi$, evaluating double 3-point finite-difference operator $\delta_z^2(\delta_z^2 q)$ gives $\omega_{\text{disc}} = \frac{16}{h^4}\nu_4$. Comparing to continuum $\omega_{\text{cont}} = \frac{\pi^4}{h^4}\nu_4$ yields $\frac{\omega_{\text{cont}}}{\omega_{\text{disc}}} = \frac{\pi^4}{16} \approx 6.08817$.

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